

Programming Assignment:

OrderedDictionary

In this assignment, you will be implementing a new Data type called `OrderedDictionary` that behaves like a dictionary except that it maintains the items in the order they were entered. (Very similar (but not exactly the same) to the `OrderedDict` datatype which is a part of the `functools` module: <https://docs.python.org/3/library/collections.html#collections.OrderedDict>)

Our custom data type has the following properties:

1. The keys of the dictionary have to be strings
2. When items are added to the dictionary, their order of insertion is remembered and when an iterator is created to iterate over the items, it should iterate in the same order as the keys were added.
3. Updating the value for a key doesn't change its order, however removing and adding back a key changes its order (on insertion it will be added at the end)
4. If the dictionary is indexed using an integer, it is treated as the index of the key insertion. So `myOrderedDictionary[0]` will give the value for the oldest key, `myOrderedDictionary[1]` will give value for the second oldest key inserted, etc.

Here's a possible way of implementing this (You can choose to implement it in your own way, as long as all the public methods are implemented correctly.):

- 1) Open the `orderreddictionary.py` file with some commented out client code in `main`.
- 2) Add a class definition for a new class called `OrderedDictionary` with
 - a. Private Attributes:
 - i. A dictionary called `__dictionary` to store the elements added.
 - ii. A list called `__listOfKeys` to store the current keys of the `__dictionary` in the order they were added. This will be used when indexing using integer values and when iterating over the dictionary.
 - b. A Constructor: Add a constructor that takes no parameters. It should initialize the attributes.
 - c. Public methods:
 - i. Special methods:

1. `__getitem__(self, index)` and `__setitem__(self, index, value)` :
These are used to access and add/update the elements of the dictionary respectively. (See the note on indexing below)
 2. `__len__(self)`
 3. `__iter__(self)`
- ii. `pop(self, index)`: Method to remove an element by an index. This method will return the value for that key.
- iii. Iterator methods similar to the dictionary data type:
1. `keys(self)`: returns an iterator to get the keys in order of insertion.
 2. `values(self)`: returns an iterator to get the values corresponding to the keys in order of insertion.
 3. `items(self)`: returns an iterator to get the (key, value) tuples corresponding to the keys in order of insertion.
- d. A note on indexing: Note that all the methods that have `index` argument (`__getitem__`, `__setitem__` and `pop`) should validate that `index` is either a string or an integer, otherwise raise a `ValueError` with an appropriate message. When `index` is a string, it should be taken as the key of the `__dictionary`. When it is an integer, it should be taken as the order of the element – so it should be used to index the `__listOfKeys` array to get the key which is then used to index the `__dictionary` to retrieve, update or remove the corresponding value as the case may be.
- 3) Uncomment the code in the main method. And run the program. Output will look like:

```
Python 3.6.3 Shell
File Edit Shell Debug Options Window Help
-----
len(d): 4
d[1]: 2
Updating d['b'] = 8...
d[1]: 8
-----
updating with integer index d[0] = 0. Should update d['a']
d['a']: 0
-----
Iterating ...
a,b,c,d,
-----
Iterating using keys()...
a,b,c,d,
-----
Removing 'b'...pop('b')
8
-----
Iterating using values()...
0,3,4,
-----
adding 'b' again d['b'] = 10
-----
Removing key 'd'...pop(2)
4
-----
Iterating using items()...
('a', 0), ('c', 3), ('b', 10),
-----
Trying d[3.0] = 5...
Invalid index type, should be str or int
-----
Trying d[5]....
list index out of range
-----
Trying pop[3.1412]....
Invalid index type, should be str or int
-----
>>> |
```

Ln: 442 Col: 4

4)

Save the file as first_last_ordereddictionary.py. Be sure to add the title comment block at the top of the file giving your name, class name, date, assignment title and description.